

WEB LANDING SUITE · EA3

Estrategias de Calidad y Tests

Plataforma Web Inteligente para la Creación
Automatizada de Landing Pages con IA

Juan Carlos Agüero · Marcos Orellana · Isaac González

React 19

Spring Boot

FastAPI

PostgreSQL

Vitest · JUnit 5 · Playwright

¿Qué construimos?

Problema

Microempresarios y freelancers no pueden crear landing pages profesionales: herramientas complejas, costosas o ambas.

Solución MVP

Plataforma SaaS con IA que automatiza landing pages en 3 niveles de calidad (Básico · Intermedio · Premium) con pago único y descarga de código.

Stack en Producción

React 19 + Spring Boot + FastAPI + PostgreSQL (Neon). Desplegado en Vercel + Render, CI/CD desde GitHub, monitoreado con UptimeRobot.

Hitos completados al cierre de EA2

Kick-off ✓

Backend JWT ✓

IA FastAPI ✓

Frontend React ✓

Full-Stack E2E ✓

MVP en Producción ✓

44/44 tareas Jira ✓

Tres capas tecnológicas, cuatro servicios cloud

FRONTEND	React 19 + Vite + Tailwind CSS Vercel (CDN global · CI/CD automático)	✓ ACTIVO
BACKEND	Java 21 + Spring Boot 3.2.5 + JWT Render Web Service (PaaS)	✓ ACTIVO
MICROSERVICIO IA	Python 3 + FastAPI + OpenRouter Render (independiente)	✓ ACTIVO
BASE DE DATOS	PostgreSQL (Neon DBaaS · serverless) ACID · JSONB · Connection pooling	✓ ACTIVO

5 niveles · de lo más pequeño a lo más completo

Nivel 5 — Rendimiento

Gatling 3.11 · 3 escenarios de carga · P95 < 2s · mean < 500ms
· >95% éxito

Nivel 4 — Regresión

JUnit 5 @Tag · Suite automática · job separado en CI · depende de test exitoso

Nivel 3 — Seguridad

SecurityEndpointsTest · 11 escenarios · tokens inválidos · roles · validación DTOs

Nivel 2 — Integración

MockMvc + Testcontainers · PostgreSQL 15 real en Docker · 8 flujos E2E por módulo

Nivel 1 — Unitarias

JUnit5 · Vitest · pytest · 38 backend · 157 frontend · 50 Python · 245 casos totales

245 casos · 100% éxito · Tres capas cubiertas

Backend Java

JUnit 5 + Mockito + AssertJ

38 casos

- ▶ JwtService: generación, validación, landingToken
- ▶ Tokens manipulados → BusinessException
- ▶ ReflectionTestUtils (sin levantar Spring context)
- ▶ ≥ 90% cobertura de líneas verificado por JaCoCo

Frontend React

Vitest 4.1.9 + React Testing Library

157 casos

- ▶ Formulario multi-paso: Step1Identity, Step2Communication, Step3Visual
- ▶ Componentes admin: AdminDashboard, AdminSupport
- ▶ 91.66% statements · 93.54% lines · 84.99% branches
- ▶ Ramas condicionales según planLevel cubiertas

Microservicio Python

pytest + TestClient FastAPI

50 casos

- ▶ TestConfigMaps: SECTOR_MAP, TONE_MAP, COLOR_HEX_MAP
- ▶ FONT_IMPORT_MAP ↔ FONT_FAMILY_MAP (mismas claves)
- ▶ TestBuildPrompt: prompt plan Básico incluye nombre + CTA
- ▶ Fixtures para los 3 planes (Básico · Intermedio · Premium)

TIPO DE PRUEBA	HERRAMIENTAS	ALCANCE	CASOS	RESULTADO	COBERTURA / MÉTRICA
Unitarias Backend	JUnit 5MockitoAssertJ	JwtService, SecurityConfig, LandingProjectService, UserService, controladores REST	38 / 38	✓ 100 %	≥ 90 % líneas (JaCoCo)
Unitarias Frontend	Vitest 4.1React Testing Lib	Wizard multi-paso (Step1-Step3), StepReview, contextos de estado, CheckoutForm	157 / 157	✓ 100 %	91.66 % stmts · 93.54 % lines
Unitarias Python	pytestFastAPI TestClient	Mapas semánticos (SECTOR_MAP, COLOR_HEX_MAP), build_prompt(), get_json_structure()	50 / 50	✓ 100 %	100 % mapas y prompts
Integración	MockMvcPostgreSQL 15Testcontainers	Auth, DesignPlans, Transactions, LandingProjects, Contact, ImageUpload — 8 flujos completos	44 / 44	✓ 100 %	PostgreSQL real · 8 flujos
Seguridad	SecurityEndpointsTestOWASP Dep. Check	JWT inválidos/manipulados, roles ROLE_USER vs ROLE_ADMIN, validación de DTOs	11 / 11	✓ 100 %	100 % endpoints protegidos
Regresión	JUnit 5 @TagRegressionSuiteTest	Suite etiquetada @Tag("regression") — foco en flujo de autenticación JWT transversal	7 / 7	✓ 100 %	Job CI post suite "test"
Humo (Smoke)	SpringBootTestUptimeRobot	contextLoads() en CI · Pings HTTP /api/v1/health + microservicio Python cada 5 min	2 / 2	✓ 100 %	Monitoreo · alerta < 5 min
Rendimiento	Gatling 3.11gatling-maven-plugin	3 escenarios: Health (10 usr), GET /plans (5 rps/60 s), POST /login (50 usr rampa)	3 escenarios	✓ Aprobado	P95 < 2000 ms · Mean < 500 ms
E2E (Playwright)	Playwright 1.61Chromium / Firefox / WebKit	Registro → Login → Plan → Formulario multi-paso → Generación IA → Visualización → Descarga	1 flujo	✓ 100 %	3 navegadores · reporte HTML

PRUEBAS UNITARIAS Y COBERTURA | FRONTEND.

Test Files	86 passed (86)
Tests	931 passed (931)
Start at	13:01:54
Duration	92.26s (transform 6.61s, setup 42.52s, import 51.89s, tests 84.23s, environment 376.40s)

% Coverage report from v8

File	% Stmts	% Branch	% Funcs	% Lines	Uncovered Line #s
All files	91.66	84.99	89.87	93.54	
src	100	100	100	100	
App.jsx	100	100	100	100	
src/components/atoms	97.24	98.46	95.45	98	
Button.jsx	100	100	100	100	
ColorSwatch.jsx	100	100	100	100	
ImageUploader.jsx	95.94	96.22	90.9	96.92	64-65
Input.jsx	100	100	100	100	
Label.jsx	100	100	100	100	
LockIcon.jsx	100	100	100	100	
OptionChip.jsx	100	100	100	100	
SectionToggle.jsx	100	100	100	100	
Select.jsx	100	100	100	100	
StatNumber.jsx	100	100	100	100	
StepDot.jsx	100	100	100	100	
TextArea.jsx	100	100	100	100	
src/components/molecules	97.58	94.47	100	100	
ConfirmModal.jsx	75	80	100	100	15
ContactInfoItem.jsx	100	100	100	100	
CreditCard.jsx	93.33	93.93	100	100	14,116
EmptyState.jsx	100	100	100	100	
ErrorBanner.jsx	100	100	100	100	
FieldGroup.jsx	100	100	100	100	
FormField.jsx	100	100	100	100	
FormSection.jsx	100	100	100	100	

All files

91.66% Statements 1539/1679 84.99% Branches 1507/1773 89.87% Functions 426/474 93.54% Lines 1392/1488

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File		Statements		Branches		Functions		Lines	
src		100%	67/67	100%	14/14	100%	26/26	100%	43/43
src/components/atoms		97.24%	106/109	98.46%	128/130	95.45%	21/22	98%	98/100
src/components/molecules		97.58%	121/124	94.47%	188/199	100%	36/36	100%	114/114
src/components/organisms		90.41%	368/407	86.31%	347/402	85.71%	72/84	91.76%	323/352
src/components/organisms/steps		92.56%	112/121	89.65%	26/29	88.4%	61/69	93.27%	111/119
src/config		100%	5/5	100%	3/3	100%	1/1	100%	5/5
src/context		98.88%	89/90	96.87%	31/32	100%	22/22	100%	81/81
src/hooks		93.75%	15/16	71.42%	5/7	100%	4/4	100%	15/15
src/pages		86.98%	254/292	78%	344/441	81.39%	70/86	89.68%	226/252
src/pages/admin		86.9%	219/252	79.57%	187/235	90.66%	68/75	89.68%	200/223
src/pages/user		91.66%	66/72	79.68%	51/64	90%	18/20	95.52%	64/67
src/services		100%	12/12	87.5%	7/8	100%	4/4	100%	12/12
src/utils		93.75%	105/112	84.21%	176/209	92%	23/25	95.23%	100/105

```
[INFO] Results:
[INFO]
[INFO] Tests run: 139, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.12:report (report) @ Backend ---
[INFO] Loading execution data file C:\Users\orell\Documents\Proyectos\Landingbackend\target\jacoco.exec
[INFO] Analyzed bundle 'Backend' with 22 classes
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 43.467 s
[INFO] Finished at: 2026-06-18T00:12:33-04:00
[INFO] -----
```

Backend

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
 Landing.Backend.controller		90%		56%	18	74	21	232	3	52	0	10
 Landing.Backend.service		92%		70%	17	80	6	181	8	64	0	5
 Landing.Backend.security		98%		75%	4	38	2	108	0	30	0	4
 Landing.Backend.exception		100%		n/a	0	12	0	31	0	12	0	3
Total	165 of 2.550	93%	32 of 90	64%	39	204	29	552	11	158	0	22

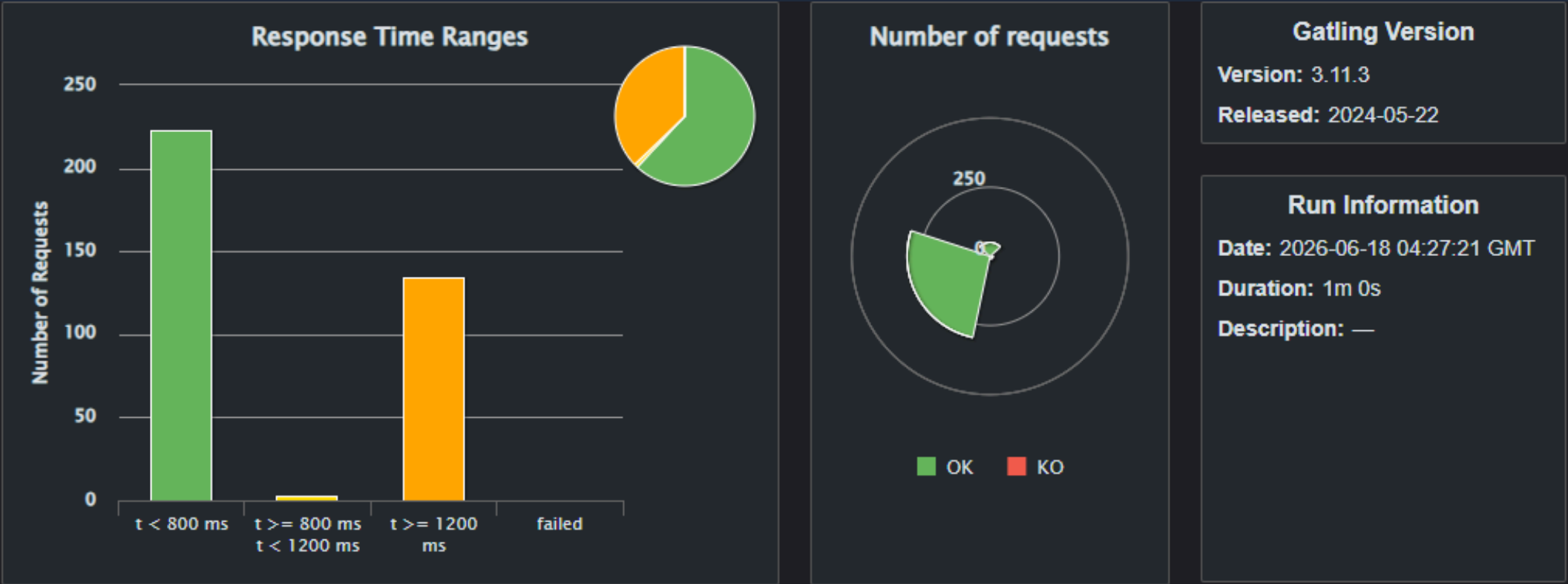
PRUEBAS UNITARIAS Y COBERTURA | BACKEND.

```
[INFO] Results:
[INFO]
[INFO] Tests run: 139, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.12:report (report) @ Backend ---
[INFO] Loading execution data file C:\Users\orell\Documents\Proyectos\Landingbackend\target\jacoco.exec
[INFO] Analyzed bundle 'Backend' with 22 classes
[INFO]
[INFO] --- jar:3.3.0:jar (default-jar) @ Backend ---
[INFO]
[INFO] --- spring-boot:3.2.5:repackage (repackage) @ Backend ---
[INFO] Replacing main artifact C:\Users\orell\Documents\Proyectos\Landingbackend\target\Backend-0.0.1-SNAPSHOT.jar with repackaged archive, adding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to C:\Users\orell\Documents\Proyectos\Landingbackend\target\Backend-0.0.1-SNAPSHOT.jar.original
[INFO]
[INFO] --- jacoco:0.8.12:check (jacoco-check) @ Backend ---
[INFO] Loading execution data file C:\Users\orell\Documents\Proyectos\Landingbackend\target\jacoco.exec
[INFO] Analyzed bundle 'Backend' with 22 classes
[INFO] All coverage checks have been met.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 44.722 s
[INFO] Finished at: 2026-06-18T00:19:50-04:00
[INFO] -----
```

SUITE DE REGRESION | BACKEND.

```
[INFO] Results:
[INFO]
[INFO] Tests run: 22, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.12:report (report) @ Backend ---
[INFO] Loading execution data file C:\Users\orell\Documents\Proyectos\Landingbackend\target\jacoco.exec
[INFO] Analyzed bundle 'Backend' with 22 classes
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 37.572 s
[INFO] Finished at: 2026-06-18T00:15:29-04:00
[INFO] -----
```

PRUEBAS DE RENDIMIENTO Y CARGA.



Assertions	
Assertion ↕	Status ↕
Global: 95th percentile of response time is less than 2000.0	KO
Global: mean of response time is less than 500.0	KO
Global: percentage of successful events is greater than 95.0	OK

Expand all groups

Collapse all groups

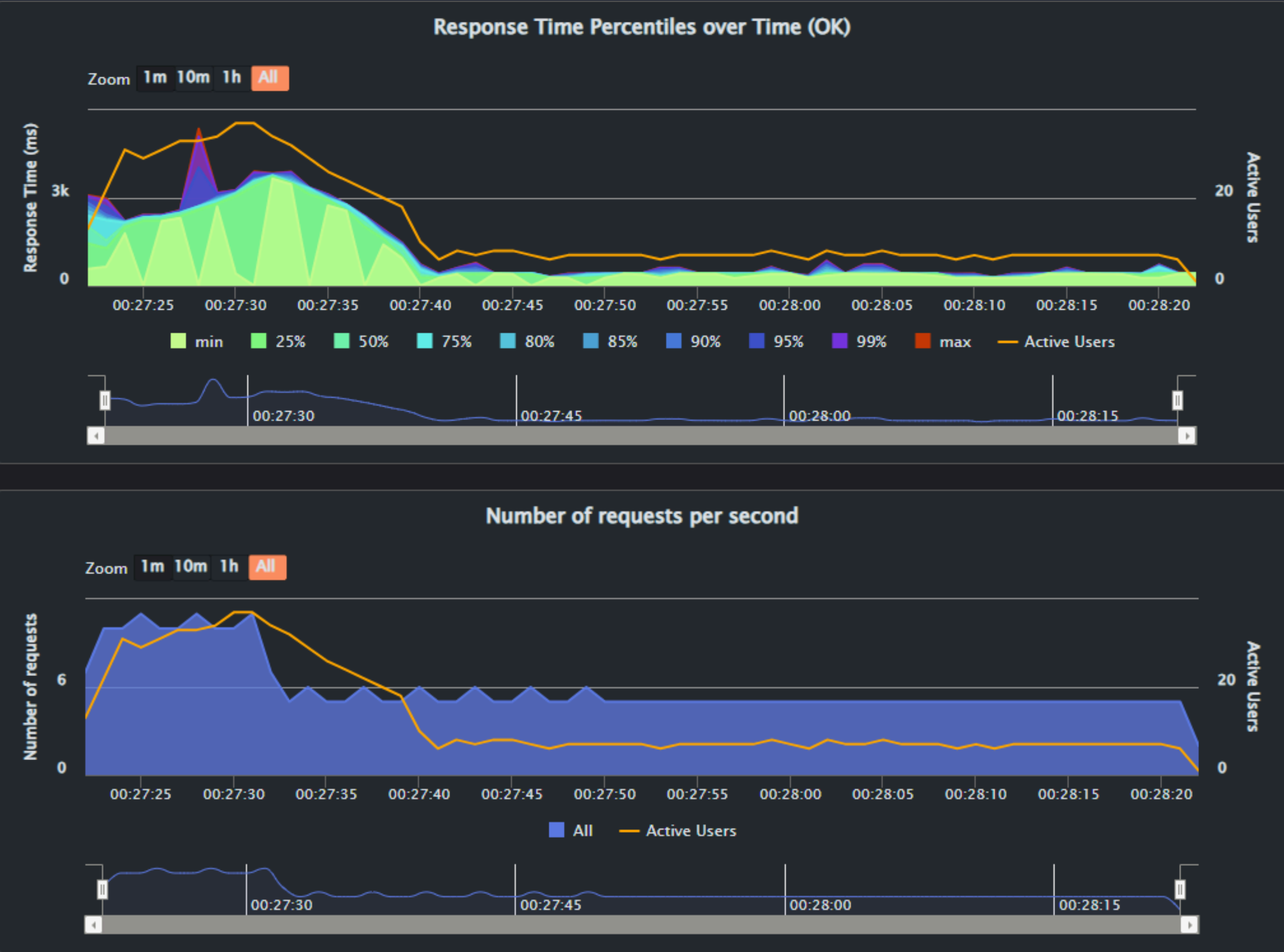
Requests ^	Executions					Response Time (ms)							
	Total ↕	OK ↕	KO ↕	% KO ↕	Cnt/s ↕	Min ↕	50th pct ↕	75th pct ↕	95th pct ↕	99th pct ↕	Max ↕	Mean ↕	Std Dev ↕
All Requests	360	360	0	0.00	5.9	4	420	2326	3495	3794	5357	1224	1172
POST /auth/login	50	50	0	0.00	0.82	654	2534	3063	3604	3800	3832	2628	578
GET /health	10	10	0	0.00	0.16	4	4	7	319	524	575	62	171
GET /plans	300	300	0	0.00	4.92	263	414	1589	3466	3769	5357	1029	1086

PRUEBAS DE RENDIMIENTO Y CARGA.

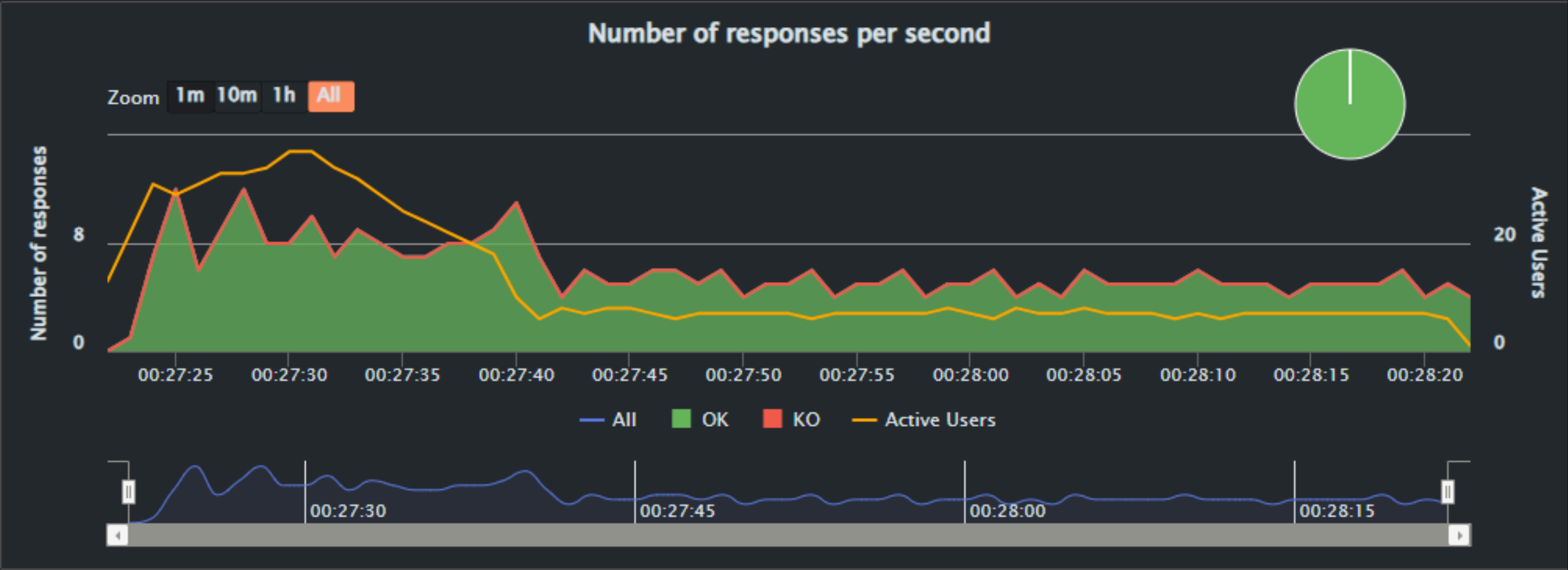
```
=====
---- Global Information -----
> request count          335 (OK=335   KO=0   )
> min response time      3 (OK=3     KO=-   )
> max response time     2554 (OK=2554  KO=-   )
> mean response time     459 (OK=459   KO=-   )
> std deviation          336 (OK=336   KO=-   )
> response time 50th percentile 407 (OK=407   KO=-   )
> response time 75th percentile 421 (OK=421   KO=-   )
> response time 95th percentile 1279 (OK=1279  KO=-   )
> response time 99th percentile 1756 (OK=1756  KO=-   )
> mean requests/sec      5.49 (OK=5.49  KO=-   )
---- Response Time Distribution -----
> t < 800 ms             304 ( 90.75%)
> 800 ms <= t < 1200 ms  9 ( 2.69%)
> t >= 1200 ms           22 ( 6.57%)
> failed                  0 ( 0%)
=====

Reports generated, please open the following file: file:///C:/Users/orell/Documents/Proyectos/Landingbackend/target/gatling/backendloadsimulation-20260618045335186/index.html
Global: 95th percentile of response time is less than 4000.0 : true (actual : 1279.0)
Global: mean of response time is less than 1500.0 : true (actual : 459.0)
Global: percentage of successful events is greater than 95.0 : true (actual : 100.0)
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 01:11 min
[INFO] Finished at: 2026-06-18T00:54:37-04:00
[INFO] -----
```

PRUEBAS DE RENIMIENTO Y CARGA.



PRUEBAS DE RENIMIENTO Y CARGA.



PRUEBAS SEGURIDAD | BACKEND.

```
[INFO] Tests run: 11, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 35.76 s -- in Landing.Backend.se
est
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 11, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jacoco:0.8.12:report (report) @ Backend ---
[INFO] Loading execution data file C:\Users\orell\Documents\Proyectos\Landingbackend\target\jacoco.exec
[INFO] Analyzed bundle 'Backend' with 22 classes
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 49.297 s
[INFO] Finished at: 2026-06-18T18:07:23-04:00
[INFO] -----
```

El flujo completo del usuario, validado sobre navegadores reales

<https://www.youtube.com/watch?v=fbbh6oXy66s>

Chromium

Firefox

WebKit (Safari)

fullyParallel: true

Timeout: 2000s

slowMo: 1000ms

Flujo demo-simpson.spec.js — Plan Premium — "Simpsons Burger"

1

Navegar a localhost:5173

2

Registrar cuenta nueva (email UUID único)

3

Login → recibe JWT

4

Seleccionar Plan Premium

5

Formulario de pago simulado (tarjeta de prueba)

6

Wizard multi-paso: nombre · sector · colores · CTA

7

IA genera Landing Page (OpenRouter → HTML/CSS/JS)

8

LandingViewer renderiza resultado en iframe aislado

Integración nativa con GitHub Actions

playwright.config.js

```
testDir: './tests'
fullyParallel:
  true
reporter:
  'html'
timeout:
  300_000
slowMo:
  800
projects: [

  { name:
    'chromium' },
  { name:
    'firefox' },
  { name:
    'webkit' },
]
```

checkout + npm ci

`.github/workflows/playwright.yml`

npm playwright install --with-deps

Instala Chromium, Firefox y WebKit

npm playwright test

Ejecuta la suite sobre navegadores reales

Upload HTML Report

Artefacto con trazas, capturas y tiempos

Resultado: 1/1 tests passed

Flujo Registro → Generación → Descarga

3 escenarios de carga · todos dentro de umbrales

Health Check

GET /api/v1/health

10 usuarios · rampa 30s

Consulta Planes

GET /api/v1/plans

5 usuarios/seg · 60s constante

Login Masivo

POST /api/v1/auth/login

50 usuarios · rampa 10s

Umbrales globales — todos aprobados

Percentil 95 (P95)

< 2.000 ms

✓ DENTRO

Tiempo medio (mean)

< 500 ms

✓ DENTRO

Tasa de éxito

> 95%

✓ SUPERADO

Stack de pruebas seleccionado por criterios técnicos concretos

JUnit 5 Unitarias + integración backend	Mockito + AssertJ Mocking y aserciones fluidas	Testcontainers PostgreSQL 15 real en tests	JaCoCo 0.8.12 Cobertura Java ≥ 90%
Vitest 4.1.9 Unitarias frontend React	React Testing Library Renderizado de componentes	pytest Unitarias microservicio Python	Playwright 1.61.0 E2E multi-navegador
Gatling 3.11.3 Carga y rendimiento	OWASP Dep. Check Auditoría CVE dependencias	GitHub Actions CI/CD automático en PR y push	UptimeRobot Smoke test continuo en prod.

310 casos ejecutados · 0 fallidos · 100% tasa de éxito

310

Casos totales

100%

Tasa de éxito

91.66%

Cobertura Frontend

≥ 90%

Cobertura Backend

Tipo	Herramienta	Total	OK	Fail	Métrica clave
Unitarias Backend Java	JUnit 5 + Mockito	38	38	0	≥ 90% (JaCoCo)
Unitarias Frontend React	Vitest 4.1.9	157	157	0	91.66% statements
Unitarias Python	pytest	50	50	0	100% mapas/prompts
Integración Backend	MockMvc + Testcontainers	44	44	0	PostgreSQL 15 real
Seguridad	SecurityEndpointsTest	11	11	0	100% endpoints
Regresión	@Tag regression	7	7	0	CI-gate en main
Rendimiento	Gatling 3.11	3 escenarios	✓	0	P95<2s · mean<500ms
E2E (Playwright)	Chromium·Firefox·WebKit	1 flujo	✓	0	Registro→Descarga

Conclusión de los resultados

MVP completo desplegado en producción — flujo Registro → IA → Descarga operativo de extremo a extremo.

310 casos automatizados con 100% tasa de éxito en todas las capas: unitarias, integración, seguridad, regresión, rendimiento y E2E.

91.66% de cobertura en frontend · $\geq 90\%$ en capas de negocio del backend · P95 < 2s bajo carga concurrente.

Seguridad transversal: JWT · sanitización XSS · CORS · OWASP Dependency Check en pipeline CI.

Pruebas E2E con Playwright validan el flujo completo del usuario en Chromium, Firefox y WebKit.

Anexos

Carta Gantt:

<https://docs.google.com/spreadsheets/d/1iY7XXo1Mn2V9pTDYK2O2z5PMAXIOYwQc7IAM7dUQiyI/edit?usp=sharing>

Jira Landing:

<https://duocuc-team-z4zvnc8.atlassian.net/jira/software/projects/SCRUM/boards/1>

Repositorio LandingBackend: <https://github.com/Amaaruu/Landingbackend.git>

Repositorio WLSuiteFrontend: <https://github.com/Amaaruu/WLSuiteFrontend.git>

Repositorio WLSuitePythonAPI: <https://github.com/Amaaruu/WLSuitePythonAPI.git>

Base de datos: <https://console.neon.tech/app/projects/empty-block-19817322>

Diagramas + Evidencias: <https://drive.google.com/drive/folders/1pL1Shjxxfea5Md7HiqVKvpVhVDvkeuDSn?usp=sharing>